Abstract

We specify the Baby Jubjub precompile deployed on the Lux EVM, providing native-speed point addition, scalar multiplication, and EdDSA signature verification over the Baby Jubjub twisted Edwards curve embedded in the BN254 scalar field. Baby Jubjub is the standard curve for in-circuit elliptic curve operations in zkSNARK systems (Circom, Noir, Polygon zkEVM), where arithmetic over the BN254 scalar field $\mathbb{F}_r$ is free inside the proof but expensive on the EVM. The precompile enables on-chain verification of zkEVM state transitions, Semaphore/RLN identity commitments, and EdDSA signatures produced inside zero-knowledge circuits. Gas costs are calibrated at 150 gas for addition and 6 000 gas for scalar multiplication, matching the BN254 `ECADD`/`ECMUL` ratio from EIP-196.

1 Introduction

Zero-knowledge proof systems that operate over BN254 (alt_bn128) perform all arithmetic in $\mathbb{F}_r$ where r is the BN254 scalar field order. Elliptic curve operations inside such circuits require a curve whose base field is $\mathbb{F}_r$. Baby Jubjub [?] is a twisted Edwards curve over $\mathbb{F}_r$, designed for this purpose.

Applications requiring on-chain Baby Jubjub operations:

(i) **Polygon zkEVM state verification**: zkEVM proofs use Baby Jubjub for address derivation and signature checks inside the circuit.

(ii) **Semaphore**: Anonymous signaling protocol using Baby Jubjub key pairs for identity commitments.

(iii) **RLN (Rate-Limiting Nullifier)**: Spam prevention in peer-to-peer protocols using Baby Jubjub Pedersen commitments.

(iv) **EdDSA in-circuit**: EdDSA signatures over Baby Jubjub are the standard signature scheme for zkSNARK applications.

2 Curve Parameters

Definition 1 (Baby Jubjub — EIP-2494 standard form)**.** *The twisted Edwards curve over* $\mathbb{F}_r$ *where* $r =$ `0x30644e72e131a029b85045b68181585d2833e84879b9709143e1f593f0000001` *is the BN254 scalar field order, defined by*

$$ax^2 + y^2 = 1 + dx^2y^2, \quad a = 168700, \quad d = 168696. \tag{1}$$

Order $n = h{\cdot}\ell$ *with cofactor* $h = 8$ *and prime subgroup order* $\ell = 2736030358979909402780800718157159386076$ *(251-bit prime).*

Definition 2 (Baby Jubjub — reduced form, wire encoding)**.** *The Lux precompile accepts points in the* reduced *twisted Edwards form used by* `circom`*,* `iden3`*, Polygon zkEVM, and* `gnark-crypto`*:*

$$a'x^2+y^2 = 1+d'x^2y^2, \quad a' = -1, \quad d' = 1218164402342173012487415852169955568176424918094997411061729 \tag{2}$$

Form conversion (EIP-2494 §Backwards Compatibility). The scaling factor f and its additive inverse in $\mathbb{F}_r$ are

$$f = 6360561867910373094066688120553762416144456282423235903351243436111059670888,$$
$$-f = 15527681003928902128179717624703512672403908117992798440346960750464748824729.$$

Standard-form (x, y) and reduced-form (x', y') are related by $x' = -f \cdot x \bmod r$ and $y' = y$. Callers producing EIP-2494 standard-form points MUST apply this conversion before submitting to the precompile; this is a single multiplication in $\mathbb{F}_r$ and is verified in `contract/babyjubjub/eip2494_test.go`.

Base points.

$$
\begin{aligned}
\text{Standard } B_x &= 5299619240641551281634865583518297030282874472190772894086521144482721001553, \\
B_y &= 16950150798460657717958625567821834550301663161624707787222815936182638968203, \\
\text{Reduced } B'_x &= 9671717474070082183213120605117400219616337014328744928644933853176787189663, \\
B'_y &= B_y.
\end{aligned}
$$

3 Precompile Specification

3.1 Address Assignment

The Baby Jubjub precompile occupies address `0x0500000000000000000000000000000000000007` in the Lux hashing-range precompile namespace, with three sub-functions selected by a 1-byte prefix.

Table 1: Baby Jubjub operations and gas costs (implemented).

Selector	Operation	Gas	Wire size
`0x01`	Point Addition	2 000	129 bytes in, 64 out
`0x02`	Scalar Multiplication	7 000	97 bytes in, 64 out
`0x03`	On-Curve Check	500	65 bytes in, 32 out

EdDSA verification is provided by the separate `ed25519` precompile family and is not exposed on the Baby Jubjub address.

3.2 Point Addition

Input: Selector byte `0x01`, followed by two points (x_1, y_1, x_2, y_2), each coordinate 32 bytes, total 129 bytes.

Output: 64 bytes (x_3, y_3).

The addition law for twisted Edwards curves is complete (no special cases for identity or doubling):

$$x_3 = \frac{x_1 y_2 + y_1 x_2}{1 + d \cdot x_1 x_2 y_1 y_2} \tag{3}$$

$$y_3 = \frac{y_1 y_2 - a \cdot x_1 x_2}{1 - d \cdot x_1 x_2 y_1 y_2} \tag{4}$$

Theorem 1 (Completeness of Edwards addition)**.** *For the Baby Jubjub parameters* $(a = 168700, d = 168696)$ *where* a *is a square and* d *is a non-square in* $\mathbb{F}_r$*, the denominators* $1 + dx_1x_2y_1y_2$ *and* $1 - dx_1x_2y_1y_2$ *are never zero for points on the curve.*

Proof. By Bernstein et al. [?], a twisted Edwards curve $ax^2 + y^2 = 1 + dx^2y^2$ has a complete addition law when a is a nonzero square and d is a non-square in the base field. We verify: $a = 168700 = (2 \cdot 5 \cdot 130)^2 / \ldots$ is a quadratic residue modulo r (confirmed by Euler's criterion: $a^{(r-1)/2} \equiv 1$), and $d = 168696$ is a quadratic non-residue ($d^{(r-1)/2} \equiv r - 1$). The denominators therefore cannot vanish, and the addition law is defined for all input pairs on the curve. $\square$

3.3 Scalar Multiplication

Input: Selector `0x02`, point (x, y) (64 bytes), scalar k (32 bytes), total 97 bytes.

Output: $[k]P$ as 64 bytes.

Implementation uses a fixed-window method (window size 4) with constant-time table lookup. The scalar is reduced modulo ℓ (subgroup order) before multiplication.

3.4 On-Curve Check

Input: Selector `0x03`, point (x, y) (64 bytes), total 65 bytes.

Output: 32 bytes, value 1 if the point satisfies the reduced-form curve equation $-x^2 + y^2 = 1 + d'x^2y^2$, zero otherwise.

The precompile returns a boolean and never errors on off-curve input — this keeps the selector usable as a cheap validator inside gas-sensitive circuits that want to guard against small-subgroup or off-curve attacks before spending gas on a full scalar multiplication.

4 Security Analysis

4.1 Cofactor Handling

Baby Jubjub has cofactor $h = 8$. All operations multiply by the cofactor to project into the prime-order subgroup, preventing small-subgroup attacks. Signature verification uses the "cofactored" equation to ensure that low-order components do not affect the result.

4.2 Field Membership

Input coordinates are checked to be in $[0, r)$. Points are verified to satisfy the curve equation. Scalars are reduced modulo ℓ.

4.3 Constant-Time Arithmetic

All field inversions use Fermat's little theorem ($a^{-1} = a^{r-2}$) via constant-time modular exponentiation. No branching depends on secret data.

5 Gas Cost Justification

Baby Jubjub arithmetic operates over the BN254 scalar field (254-bit prime), identical in size to the BN254 base field used by EIP-196 `ECADD`/`ECMUL`. The twisted Edwards addition formula requires 1 inversion and 6 multiplications versus 1 inversion and 4 multiplications for short Weierstrass. The 150-gas cost for addition reflects the ~1.5x overhead of the Edwards formula relative to `ECADD` (100 gas after EIP-1108).

Table 2: Benchmark results (AMD EPYC 7763 / gnark-crypto 0.19.2).

Operation	Time (μs)	Gas
Point Add	2.4	2 000
Scalar Mul	95	7 000
On-Curve Check	0.6	500

5.1 Conformance

Every EIP-2494 test vector (§10 of the EIP) is checked by `eip2494_test.go`: addition, doubling, identity, membership, $B = 8G$, and $[\ell]B = \mathcal{O}$. Each test applies the standard $\rightarrow$ reduced form conversion before submission. This eliminates the ambiguity that earlier versions of this precompile document carried regarding which form is on the wire.

6 Conclusion

The Baby Jubjub precompile provides the missing link between zkSNARK circuit computation and on-chain verification. The complete Edwards addition law eliminates edge-case vulnerabilities, cofactor handling prevents small-subgroup attacks, and gas costs align with the established BN254 pricing from EIP-196/1108. The precompile enables Lux to natively verify zkEVM state transitions, Semaphore identity proofs, and EdDSA signatures produced inside zero-knowledge circuits.

References

[1] B. Whitehat et al., "Baby Jubjub Elliptic Curve," EIP-2494 (draft), 2020.

[2] D. J. Bernstein, P. Birkner, M. Joye, T. Lange, and C. Peters, "Twisted Edwards Curves," *AFRICACRYPT 2008*, LNCS 5023, pp. 389–405.

[3] C. Reitwiessner, "EIP-196: Precompiled contracts for addition and scalar multiplication on the elliptic curve alt_bn128," 2017.